

40 mins quiz on 24th August.

4-4:40pm.

5% of the course.

(-25/100) penalty for cheating. (If your final marks at the end of the course is 50/100 then ————— 25/100).



$$\pi_J(\theta) \propto \sqrt{I(\theta)}$$
$$\phi = h(\theta)$$
$$\pi_J(\phi) \propto \sqrt{I(\phi)}$$
$$\phi = h(\theta)$$
$$I(\phi) = I(\theta) \left| \frac{d\theta}{d\phi} \right|$$
$$E \left( \frac{\partial^2 \log L}{\partial \theta^2} \right) \Big|_{\theta}$$

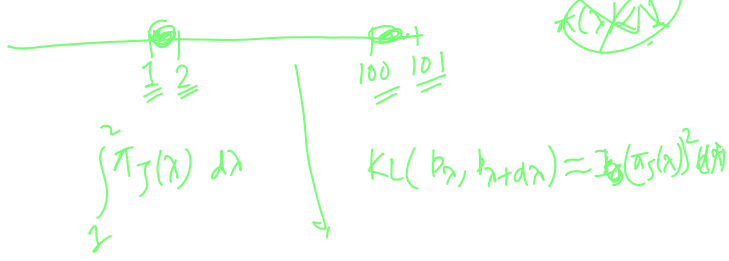


$$\text{Bin}(n, p)$$
$$\text{Var}(T(X)) \geq \frac{1}{I(\theta)}$$
$$\text{Var}(T(X)) \geq \frac{p(1-p)}{n}$$
$$T(X) = \bar{X}$$
$$I(\theta)$$

$$\text{Poisson}(\lambda)$$
$$\lambda \in \mathbb{R}^+$$

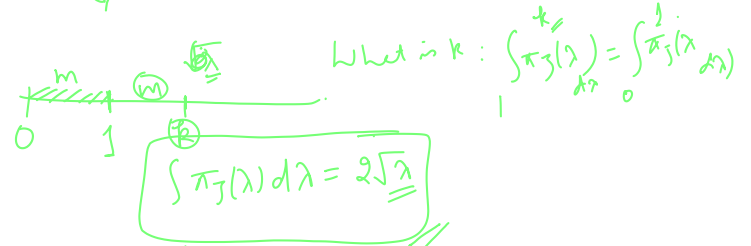
① Derive Jeffreys's prior for Poisson( $\lambda$ ).

$$\pi_J(\lambda) \propto \frac{1}{\sqrt{\lambda}}$$



② Derive for Binomial.

$$q = \log \frac{p}{1-p}$$



$N(\mu, \sigma^2)$   $\sigma^2$  known.

$$\pi_J(\mu) \propto 1$$

$$X = Z + \bar{M} \text{ where } Z \sim N(0, \sigma^2)$$

Bin:  $\pi_J(p) \propto \text{Beta}(\frac{1}{2}, \frac{1}{2})$ .

Poisson:  $\pi_J(\lambda) \propto \text{Gamma}(\frac{1}{2}, 0)$

Normal: Known  $\sigma^2$ :  $\pi_J(\mu) \propto 1$ .  
Unknown  $\sigma^2$ :  $\pi_J(\mu, \sigma^2) \propto \frac{1}{\sigma^3} \exp(-\frac{1}{2\sigma^2})$



$$\pi(\mu) < 1$$

$$100_{km} - 250_{km}$$

Non-informative prior: Jeffreys's prior.

Flat prior:  $\pi(\theta) \propto 1$ .

Diffuse/Vague prior:  $N(\mu_0, \sigma_0^2)$  where  $(\sigma_0^2) \rightarrow \infty$ .

Normal mean known, variance unknown  $\pi_J(\sigma^2) \propto \frac{1}{\sigma^3}$ .

Multi-parameter models

$$2.5 \ln^{-1}(\sqrt{b})$$



# Multi-parameter models.

$$(\theta_1, \theta_2) \quad \pi(\theta_1, \theta_2) \quad \downarrow \text{prior}$$

$$x_1, \dots, x_n \sim_{\theta_1, \theta_2} f(x; \theta_1, \theta_2)$$

e.g.  $\theta_1 = \mu, \theta_2 = \sigma^2$ .

$$x_1, \dots, x_n \sim_{\mu, \sigma^2} N(\mu, \sigma^2)$$

$$\pi(\theta_1, \theta_2 | D) \propto \pi(\theta_1, \theta_2) \pi(D | \theta_1, \theta_2) \quad \pi(\mu, \sigma^2) \checkmark$$

$\mu, \sigma^2$  nuisance parameters.

$$\pi(\theta_1 | D) = \int \pi(\theta_1, \theta_2 | D) d\theta_2$$

$$(X, Y) \quad X^{(1)}, Y^{(1)} \quad X^{(2)}, Y^{(2)} \quad \dots \quad X^{(B)}, Y^{(B)}$$

$$\pi(X, Y) = \pi(X | Y) \pi(Y)$$

$$\pi(\theta_1 | D) = \int \pi(\theta_1, \theta_2 | D) d\theta_2 = \int \pi(\theta_1 | \theta_2, D) p(\theta_2 | D) d\theta_2$$

$$\pi(\theta_1 | \theta_2, D) \quad \pi(\theta_2 | D)$$

Single-parameter inference.

$$\pi(\theta_1 | D) \quad \text{Monte-Carlo integration}$$

$$\theta_1^{(1)}, \theta_2^{(1)} \quad \dots \quad \theta_1^{(B)}, \theta_2^{(B)}$$

Marginal of  $\theta_1 | D$  can be obtained by ignoring the samples of  $\theta_2$

$$\theta_2 \rightarrow M_1, M_2 \quad \pi(\theta_1 | M, D)$$

Suppose you want to estimate  $\mu$  &  $\sigma^2$  in  $N(\mu, \sigma^2)$  model.

For known value of  $\sigma^2$  you know how to estimate.

and you also know.  $p(\sigma^2 | D)$

$$\pi(\mu | D) = \int \pi(\mu | \sigma^2, D) \pi(\sigma^2 | D) d\sigma^2$$

$$\pi(\mu, \sigma^2)$$

$\sigma^2$  (known) this needs to be calculated

Generating sample from  $\pi(\mu | D)$  can be done in 2-steps:

Step 1: Generate samples from  $\sigma^2 | D$  (call it  $\sigma^{2(s)}$ )

Step 2:  $\mu | \sigma^{2(s)}, D$

Gibbs sampling:  $\sigma^2 | \mu, D \rightarrow \sigma^{(s)}$  conditional in Gibbs sampler.  
 $\mu | \sigma^2, D \rightarrow \mu^{(s)}$  conditional in Gibbs sampler.  
 $\mu^{(s+1)} | \sigma^{2(s+1)}, D$  Dependence.

Go to step 1 & Iterate

$$(X, Y) \quad \pi(X) \rightarrow \text{generate from marginal dist'n of } X$$

& then  $\pi(Y | X) \rightarrow$  conditional distribution

Repeat

Scaled-inverse- $\chi^2$  distribution:

$$f(x; \nu, \tau^2) = \frac{\tau^\nu (\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \left( \frac{\nu}{2} + 1 \right) e^{-\frac{\nu \tau^2}{2x}}, \quad x > 0$$

Scaled-inv- $\chi^2(\nu, \tau^2)$ .  $\nu$  is degrees of freedom.  $\tau^2$  is scale

Normal unknown  $\sigma^2$ , known  $\mu$ .

If  $X \sim \text{Scaled-inv-}\chi^2(\nu, \tau^2)$ .

then a)  $\frac{\nu \tau^2}{X} \sim \chi^2(\nu)$  b)  $X \sim \text{inv-gamma}(\frac{\nu}{2}, \frac{\nu \tau^2}{2})$

$$c) kX \sim \text{Scaled-inv-}\chi^2(\nu, k\tau^2) \quad \left[ \frac{\nu \tau^2 k}{kX} \sim \chi^2(\nu) \Rightarrow kX \sim \text{Scaled-inv-}\chi^2(\nu, k\tau^2) \right]$$

Let  $X_1, \dots, X_n \sim N(\mu, \sigma^2)$   $\mu$  is known.

Let  $\sigma^2 \sim \text{Scaled-inverse-}\chi^2(\nu_0, \sigma_0^2)$ .

Then,  $\sigma^2 | D \sim \text{Scaled-inverse-}\chi^2(\nu, \tau^2)$  where  $\nu = \nu_0 + n$ ,  $\tau^2 = \frac{\nu_0 \sigma_0^2 + \sum (X_i - \mu)^2}{\nu}$ .

$\sigma^2 \sim \text{Scaled-inverse-}\chi^2(\nu_0, \sigma_0^2)$   
 $y|\sigma^2 \sim N(\mu, \sigma^2)$   $\mu$  is known

Then, find the marginal dist<sup>n</sup> of  $y$ .

$y \sim t_{\nu_0}(\mu, \sigma_0^2)$  → DO IT

⑦  
 $\sqrt{2}$

$p(y) = \int p(y|\sigma^2) \pi(\sigma^2) d\sigma^2$   
 $y|\sigma^2 \sim N(\mu, \sigma^2)$

$\sigma^2 \rightarrow$  known ✓  
 $\mu \rightarrow$  unknown &  $\mu$  follows normal

↓  
 Marginal of  $y$  is normal.

$\sigma^2|D \sim \text{Scaled-inverse-}\chi^2(\nu_n, s_n^2)$

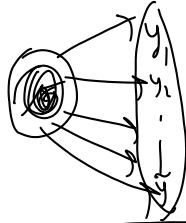
⑧  $p(\tilde{y}|D)$  (Posterior predictive distribution).

D.  
 $y_1, \dots, y_n \sim N(\mu, \sigma^2)$   $\mu$  known.

$\sigma^2 \sim \text{Scaled-inverse-}\chi^2(\nu_0, \sigma_0^2)$

$\sigma^2|D \sim \text{Scaled-inverse-}\chi^2(\nu_n, s_n^2)$

$y|\sigma^2, D$



$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$  → Find.  ~~$p(\mu, \sigma^2)$~~

$y_1, \dots, y_n \sim N(\mu, \sigma^2)$  → Scaled-inverse- $\chi^2$   ~~$p(\sigma^2|y)$~~   
 $p(\sigma^2|y)?$